



RAJALAKSHMI
INSTITUTE OF
TECHNOLOGY

ME23312 ENGINEERING THERMODYNAMICS

Unit I – Basics, Zeroth and First Law · Presentation 6 of 10

Displacement Work and the p–V Diagram

Work of a moving boundary · area under the p–V curve · work as a path function · the polytropic family · work in the standard processes

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In this presentation

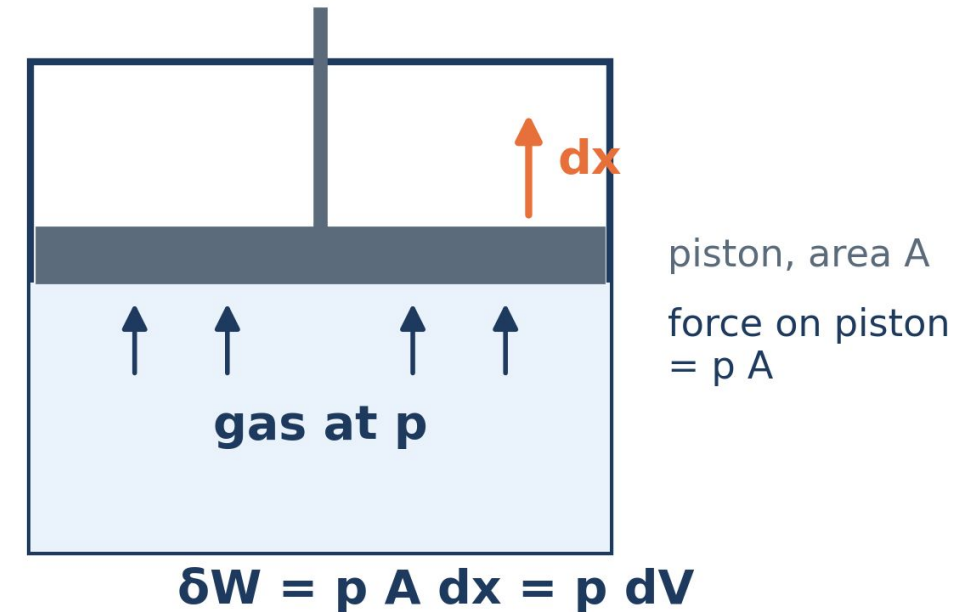
1. Work of a moving boundary
 2. Work as the area under the p–V curve
 3. Conditions for $W = \int p \, dV$
 4. Work is a path function
 5. The polytropic family $pV^n = C$
1. Constant-pressure and constant-volume processes
 2. Isothermal process
 3. Polytropic and reversible adiabatic processes
 4. Summary table of work expressions
 5. What the p–V diagram does not show

1

Work of a moving boundary

Work of a moving boundary

- Gas at pressure p in a cylinder closed by a frictionless piston of area A
- Quasi-static process: the gas pressure acts uniformly on the piston
- Piston moves an infinitesimal distance dx : force pA does work $\delta W = pA dx$
- $A dx$ is the change in volume, so $\delta W = p dV$

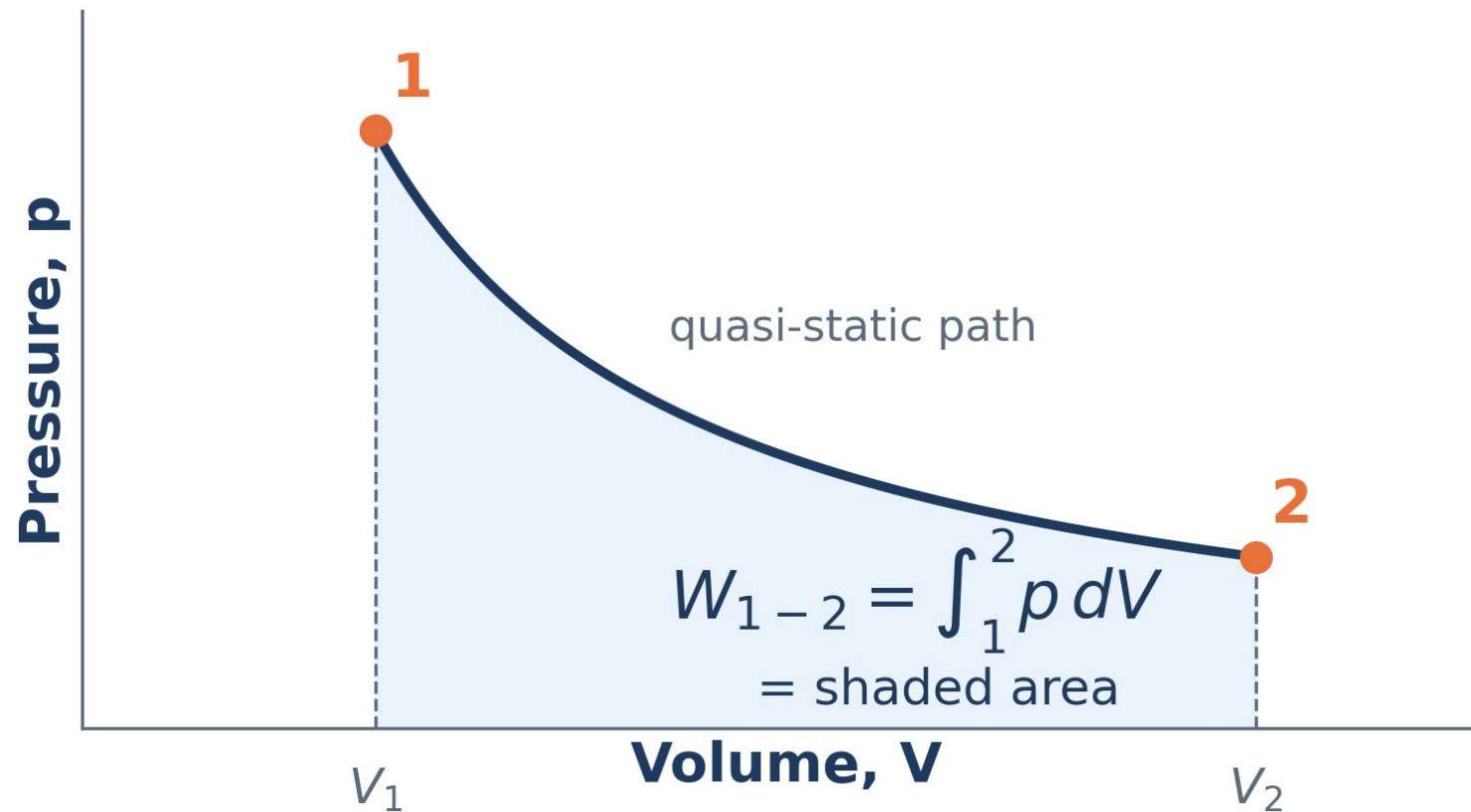


Displacement ($p \, dV$) work

$$W_{1-2} = \int p \, dV = \text{area under the process curve on the } p\text{--}V \text{ diagram}$$

- Positive for expansion ($dV > 0$) – work done **by** the gas
- Negative for compression ($dV < 0$) – work done **on** the gas

Work as the area under the p–V curve



Quasi-static expansion 1 → 2: the shaded area is the work done by the gas

Three conditions for $W = \int p \, dV$

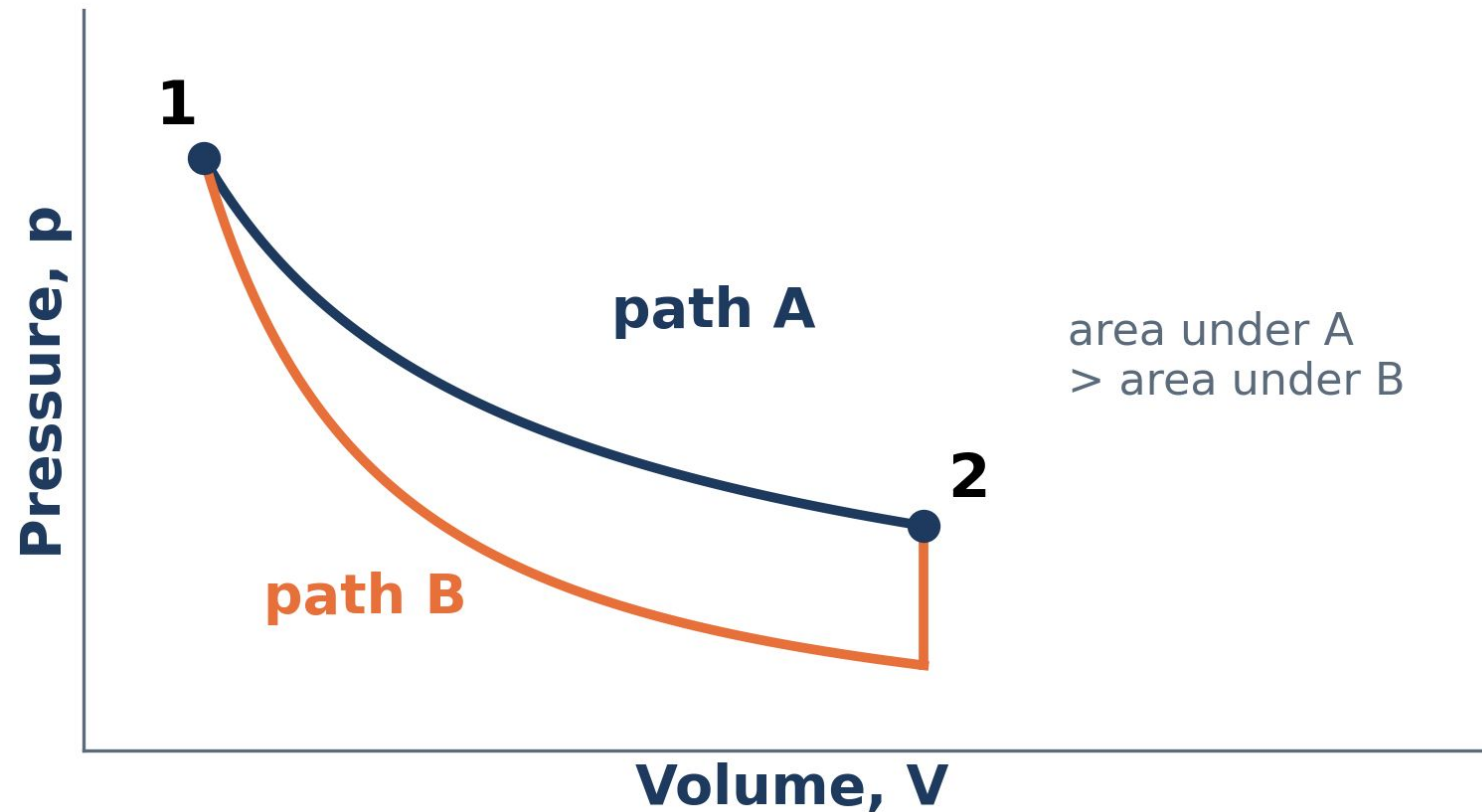
1. The process must be **quasi-static**, so that p is defined and uniform
2. The boundary must **move** ($dV \neq 0$)
3. p must be the pressure of the **system**, not of the surroundings

KEY POINT Free expansion violates the first condition – the work is zero, whatever the volume change.

2

Work is a path function

Two paths between the same end states



Same end states 1 and 2, different paths – area under A > area under B

Why work is a path function

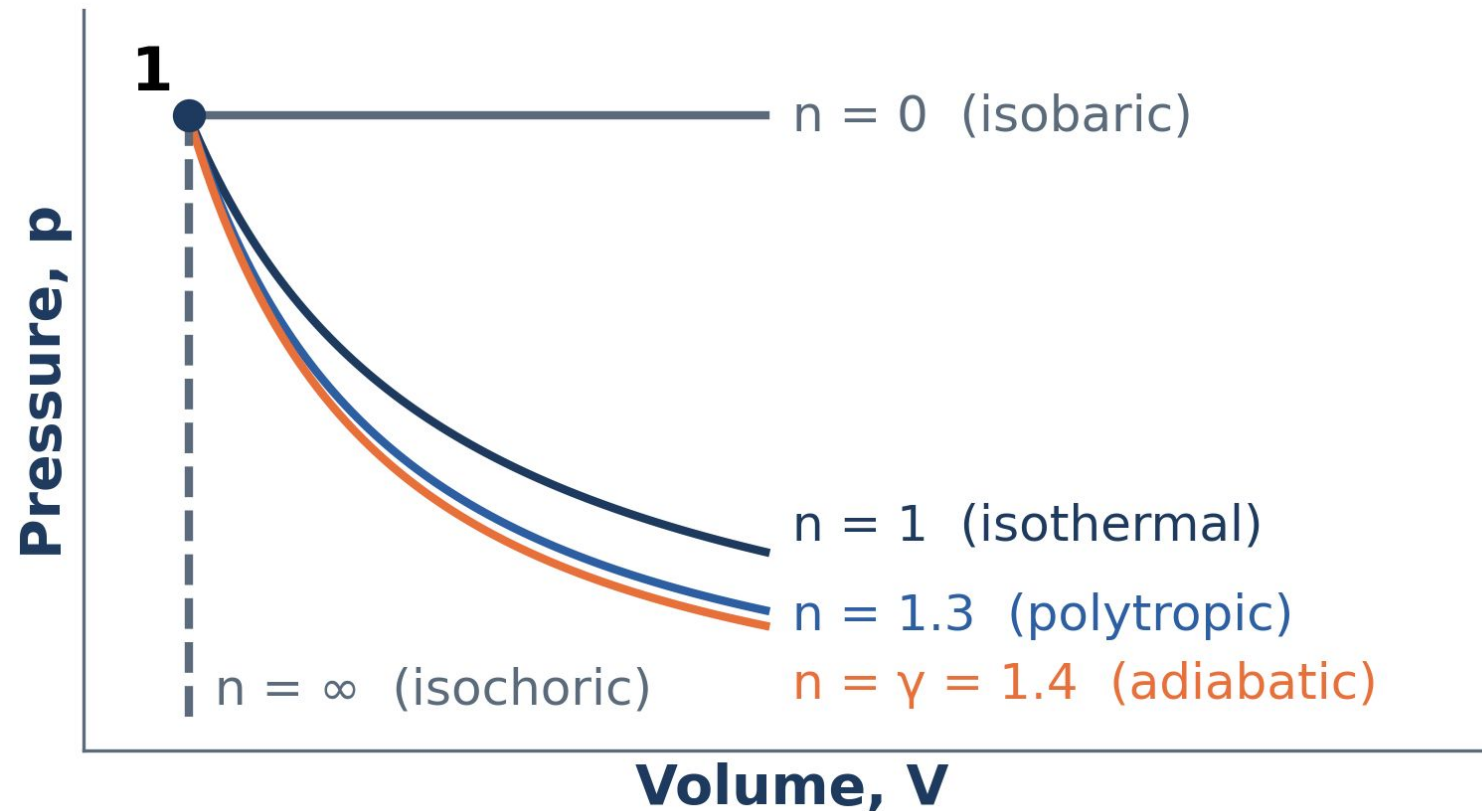
- The area under the curve depends on the curve, not merely on its end points
- Different paths between the same states give different work transfers
- δW is an **inexact differential**; it integrates to W_{1-2} , never to $W_2 - W_1$
- For a cycle $\oint \delta W \neq 0$: the **enclosed area** is the net work of the cycle

3

Work in the standard processes

for a fixed mass of ideal gas, $pV = mRT$

The polytropic family $pV^n = C$



$n = 0$ isobaric · $n = 1$ isothermal · $n = \gamma$ reversible adiabatic · $n = \infty$ isochoric

The polytropic index n

- $n = 0$: constant pressure (isobaric)
- $n = 1$: isothermal, $pV = C$
- $n = \gamma = c_p/c_v$: reversible adiabatic, $pV^\gamma = C$
- $n = \infty$: constant volume (isochoric)
- Real compressions and expansions: typically $n = 1.2$ to 1.35

Constant pressure and constant volume

Constant pressure ($n = 0$): $W_{1-2} = p (V_2 - V_1) = mR (T_2 - T_1)$

Constant volume ($n = \infty$): $W_{1-2} = 0$

- At constant volume $dV = 0$, so there is no displacement work
- Paddle or electrical work may still be done on the gas, but it does not appear as an area

Isothermal process, $pV = C$ ($n = 1$)

$$p = C/V \Rightarrow W_{1-2} = C \int dV/V$$

$$W_{1-2} = p_1 V_1 \ln (V_2/V_1) = mRT \ln (V_2/V_1) = p_1 V_1 \ln (p_1/p_2)$$

- For an ideal gas the temperature, and therefore pV , stays constant
- The curve is a rectangular hyperbola on the p – V diagram

Polytropic process, $pV^n = C$ ($n \neq 1$)

$$p = C V^{-n} \Rightarrow W_{1-2} = C \int V^{-n} dV$$

$$W_{1-2} = (p_1 V_1 - p_2 V_2)/(n - 1) = mR (T_1 - T_2)/(n - 1)$$

- $T_2/T_1 = (V_1/V_2)^{n-1} = (p_2/p_1)^{(n-1)/n}$
- Index from two known states: $n = \ln (p_2/p_1) / \ln (V_1/V_2)$

Reversible adiabatic process, $pV^\gamma = C$

$$W_{1-2} = (p_1 V_1 - p_2 V_2)/(\gamma - 1)$$

$$= mR (T_1 - T_2)/(\gamma - 1) = mc_v (T_1 - T_2)$$

- Same as the polytropic result with $n = \gamma$
- Also follows from the first law with $Q = 0$: $W = -\Delta U$
- Steeper than the isothermal curve on the p – V diagram

Displacement work – summary

Process	Relation	Work W_{1-2}
Constant pressure	$p = C$	$p (V_2 - V_1)$
Constant volume	$V = C$	0
Isothermal (ideal gas)	$pV = C$	$p_1 V_1 \ln (V_2/V_1)$
Polytropic	$pV^n = C$	$(p_1 V_1 - p_2 V_2)/(n - 1)$
Reversible adiabatic	$pV^\gamma = C$	$(p_1 V_1 - p_2 V_2)/(\gamma - 1)$

What the p - V diagram does not show

The p - V diagram shows displacement work only, and only for quasi-static processes.

- Work done by a paddle, a shaft or an electric heater does not appear as an area
- In a free expansion the area under the (undefined) path is *not* the work – the work is zero

Summary

- $\delta W = p \, dV$ for a quasi-static moving boundary; $W_{1-2} = \int p \, dV$
- Work = area under the p – V curve; positive for expansion
- Work is a path function – different paths, different areas; $\oint \delta W =$ enclosed area
- Polytropic family $pV^n = C$: $n = 0, 1, \gamma, \infty$
- Isobaric $p(V_2 - V_1)$; isochoric 0; isothermal $p_1 V_1 \ln(V_2/V_1)$
- Polytropic $(p_1 V_1 - p_2 V_2)/(n - 1)$; adiabatic – the same with $n = \gamma$
- p – V area shows displacement work only – not paddle, shaft or electrical work

Next: The first law for closed systems

- Joule's experiments and the first law for a cycle
- Energy is a property: $Q - W = \Delta E$
- Internal energy, the PMM1 and the limitations of the first law
- Numerical problems: Worked Examples 1.2 – 1.5 in the lecture notes